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Jul 28, 2025 at 6:06 PM

explain the three tier architecture of

The **Three-Tier Architecture of DBMS (Database Management System)** is a client-server architecture that separates the database system into three layers or tiers:

1. Presentation Tier (Client Tier / User Interface Layer):

- **Who interacts here?** → End Users
- **What happens here?** → This layer is responsible for user interaction. It presents data to the user and takes user input.
- **Examples:** Web browsers, mobile apps, desktop applications.
- **Characteristics:**
 - No direct database logic.
 - Only handles the display (GUI or Web UI).
 - Communicates with the application tier (middle tier).

2. Application Tier (Middle Tier / Business Logic Layer):

- **Who interacts here?** → Application Server
- **What happens here?** → It acts as a mediator between the client and the database.
 - Contains business logic and data processing rules.
 - Receives input from the client layer, processes it, and interacts with the database tier.
- **Examples:** Web servers, API services, application servers (like Apache Tomcat, Flask, Django, etc.).
- **Characteristics:**
 - Ensures data security and validation.
 - Enforces business rules.
 - Handles application logic and workflow.

3. Data Tier (Database Tier / Data Storage Layer):

- **Who interacts here?** → Database Server
- **What happens here?** → This layer is where the actual data resides.
 - Responsible for storing, retrieving, and managing data.
- **Examples:** Oracle, MySQL, PostgreSQL, MongoDB, etc.
- **Characteristics:**
 - Handles query processing and data management.
 - Only interacts with the Application Tier.
 - Implements security policies, backups, and data integrity constraints.

Visualization Diagram:



Advantages of Three-Tier Architecture:

1. **Separation of Concerns** → Each layer handles its specific responsibility.
2. **Scalability** → Can scale UI, business logic, and database independently.
3. **Maintainability** → Easier to modify or update one layer without affecting others.
4. **Security** → Direct access to data is prevented; only the application tier talks to the DB.
5. **Flexibility** → Different technologies can be used at each layer.

Example Scenario:

- **User** fills a form on a website (**Presentation Tier**).
- The form data is sent to a **web server (Application Tier)** which applies business rules.
- The server queries the **Database (Data Tier)** to store/retrieve data.
- The response goes back up the layers to display on the user's screen.

Do you want me to also explain **Two-Tier Architecture vs Three-Tier Architecture comparison?**

ChatGPT can make mistakes. Check important info.